A finite hyperbolic group with nonzero Whitehead group

Disproof of the unrestricted vanishing clause in TLMC 00000001520

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Exact scope. The English source says: "For Gromov-hyperbolic groups Γ, $\mathrm{Wh}(\Gamma) = 0$ (torsion-free and trivial)." The Chinese likewise places its parenthesis after the assertion $\mathrm{Wh}(\Gamma) = 0$. Neither version imposes torsion-freeness on Γ. We refute that printed unrestricted vanishing assertion. The counterexample has torsion, so it does not refute a corrected assertion restricted to torsion-free groups. We make no claim about the subsequent boundary-action, 2-torsion, or Farrell–Jones clauses. The complete bilingual source is in `SOURCE.md`.

Theorem 1. *Let $G = C_5 = \langle t \mid t^5 = 1\rangle$ and $A = \mathbb{Z}[G]$. Then G is a finitely generated Gromov-hyperbolic group and $\mathrm{Wh}(G) \neq 0$. More precisely, the genuine group-ring unit*

$$u = 1 - t - t^{-1}, \qquad v = 1 - t^2 - t^{-2}$$

satisfies $uv = vu = 1$ and has nonidentity image in $\mathrm{Wh}(G)$.

1. The actual stable groups and Whitehead quotient. For an arbitrary commutative ring R, take finite ordered index sets $I_0 = \{*\}$, $I_{n+1} = I_n \sqcup \{*\}$. Their proved cardinalities are $n + 1$. The maps

$$\mathrm{GL}(I_n, R) \longrightarrow \mathrm{GL}(I_{n+1}, R), \qquad M \longmapsto \begin{pmatrix} M & 0 \\ 0 & 1 \end{pmatrix}$$

are actual injective homomorphisms. Lean constructs the whole $\mathbb{N}$-indexed functor and its native filtered colimit $\mathrm{GL}_\infty(R)$ in the category of groups. We use the standard definition $K_1(R) = \mathrm{GL}_\infty(R)^{\mathrm{ab}}$, realized by native abelianization. This is the stable-abelianization definition of algebraic K_1; the equivalence with the elementary-matrix definition is Whitehead's lemma, recalled in Milnor [1, pp. 358–360].

For $A = \mathbb{Z}[C_5]$, let $T \leq A^\times$ be the range of the true homomorphism

$$\mathbb{Z}^\times \times C_5 \longrightarrow A^\times, \qquad (\epsilon, g) \mapsto \epsilon g.$$

The classification $\mathbb{Z}^\times = \{1, -1\}$ is used in Lean to prove that T is exactly the subgroup generated by -1 and the group elements. If $j : A^\times \to K_1(A)$ is the actual one-by-one-matrix inclusion, the formal Whitehead group is

$$\mathrm{Wh}(C_5) = K_1(A)/j(T).$$

Lean also proves that $j(T)$ is the subgroup generated by the images of -1 and all group elements. This is the signed quotient of the standard definition in Milnor [1, p. 373].

2. A determinant detector and a concrete nonzero class. The ordinary determinants commute with every block inclusion. Their cocone gives, by the native group-colimit universal property, $\det_\infty : \mathrm{GL}_\infty(A) \to A^\times$. Native abelianization and quotient constructions then give genuine homomorphisms

$$K_1(A) \longrightarrow A^\times, \qquad \mathrm{Wh}(C_5) \longrightarrow A^\times/T.$$

The first map sends $j(a)$ to a, and the second sends the Whitehead class of a unit a to its actual coset aT. The unit quotient is only a detection target; it is not substituted for K_1 or Wh.

In the actual group algebra, $t^{-1} = t^4$ and $t^{-2} = t^3$. Hence

$$(1 - t - t^4)(1 - t^2 - t^3) = 1 + (t + t^2)(t^5 - 1) = 1.$$

This proves invertibility with the displayed inverse. The coefficients of u at the two distinct group elements t and t^{-1} are both -1. Every signed group unit ϵg is supported at one group element. Consequently $u \notin T$. Its coset in $A^\times/T$ is nonidentity, so the determinant detector proves that its actual Whitehead class is nonidentity as well.

Milnor's Example 6.6 uses $t + t^{-1} - 1$; our u is its negative and therefore has the same signed Whitehead class. The argument above needs neither the classification $\mathrm{Wh}(C_5) \cong \mathbb{Z}$ nor a claim that SK_1 vanishes. In particular, it does not use Milnor's general finite-abelian-group statement in Example 1.6, whose added-in-proof warning must be respected.

3. The genuine Cayley metric and hyperbolicity. Use the finite symmetric generating set $S = C_5 \setminus \{1\}$. Lean proves $|C_5| = 5$, $|S| = 4$, and $\langle S \rangle = C_5$ from the actual finite group. Its native Cayley graph is the complete graph on these five vertices and is connected. Define d as the least length of a Cayley walk, using `SimpleGraph.dist`. Existence of a shortest walk is proved before the distance is used, so the disconnected-distance default is never invoked. One obtains

$$d(g, h) = \begin{cases} 0 & g = h, \\ 1 & g \neq h. \end{cases}$$

Lean equips the group with the resulting real-valued native `MetricSpace`, proves left invariance, and verifies the standard four-point inequality, for all actual group elements,

$$d(a, c) + d(b, d) \leq \max\{d(a, b) + d(c, d), d(a, d) + d(b, c)\} + 2.$$

Indeed the left side is at most 2, while the maximum is nonnegative. This is the four-point formulation of hyperbolicity for the word metric. The standard relationship with the geodesic Cayley-graph formulation is described in Bowditch [2, §§2.3,6.5,6.10]; his §6.10 explicitly includes every finite group. The development proves the full Cayley-walk metric statement directly, rather than attaching a hyperbolicity label to an unrelated group. It does not formalize the general equivalence of four-point and thin-triangle definitions. The proved element $t \neq 1$ with $t^5 = 1$ records exactly why this counterexample is outside a torsion-free-group revision.

Formalization and reproduction. The final theorem is `TLMC1520.complete_counterexample`. The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

From `lean/`, run these four commands separately:

```
lean --version
lake build +Main
lake env lean -DwarningAsError=true Check.lean
lake env leanchecker --fresh Main
```

The audit emits ordinary `#print axioms` for every actual owned declaration, including generated names. Only `propext`, `Classical.choice` and `Quot.sound` are permitted. Source-bound logs, execution receipts, actual thread samples and a SHA-256 manifest accompany the submission. Production timings are diagnostic and give no controlled efficiency conclusion.

References

[1] J. Milnor, *Whitehead torsion*, Bulletin of the American Mathematical Society 72 (1966), 358–426, §1, §6, Example 6.6. DOI: 10.1090/S0002-9904-1966-11484-2.

[2] B. H. Bowditch, *A course on geometric group theory*, author notes, October 2005, §2.3 (p. 12), §6.5 (pp. 51–53), §6.10 (p. 59). Author PDF.